

Project Report on  
**Efficient Multi-class Classification of YouTube Texts using  
Pre-trained Transformer Models**

Submitted by  
Moyaj Ibne Akbar  
ID: ASH2022044M  
Session: 2019-20

Submitted to  
Md. Shahriare Satu  
Assitant Professor



Department of Management Information Systems  
Noakhali Science and Technology University  
Sonapur, Noakhali-3814

Date of Submission: May 30, 2024

# TABLE OF CONTENTS

## CHAPTER

|                                                  |    |
|--------------------------------------------------|----|
| <b>I. Introduction</b>                           | 1  |
| 1.1 Overview                                     | 1  |
| 1.2 Major Contributions                          | 2  |
| <b>II. Methods and materials</b>                 | 3  |
| 2.1 Proposed Methodology                         | 3  |
| 2.2 Model used in Text Classification            | 4  |
| 2.3 Performance Metrics                          | 5  |
| <b>III. Experiments results &amp; Discussion</b> | 8  |
| 3.1 Dataset Descriptions                         | 8  |
| 3.2 Experimental Setup                           | 9  |
| 3.3 Discussion                                   | 10 |
| <b>References</b>                                | 11 |

# CHAPTER I

## Introduction

### 1.1 Overview

Video content on the web platform has increased explosively during the past decade, thanks to the open access to Facebook, YouTube, etc. YouTube, founded in early 2005, has grown rapidly to become the second largest video sharing platform with approximately 37 million channels [7]. YouTube is a video sharing website that allows users to watch, share, enjoy, and comment on, and upload their videos. This implies that people are constantly looking for information on YouTube and discovering videos on different topics of interest. The YouTube platform serves as an easy way for users to share and save videos online. Every minute, individuals all around the world post approximately 300 hours of video on YouTube. YouTube videos cover a large variety of subjects that someone chooses to advertise a video on. These videos are easy to distribute via social media platforms, websites, and email, and they may be even embedded in other websites. YouTube video content categories manage YouTube channels and videos. The finestsuited category for video aids the viewers in easily finding the video. To gain better visibility and rank, videos must be categorized, and different tags must be used. The availability of such a large amount of video content requires automatic categorization to make it easily searchable and useful for the users [7].

Deep learning models have shown exceptional performance across numerous domains, particularly in image and video classification tasks [9]. In recent years, Natural Language Processing (NLP) has seen significant advancements through deep learning, especially using word-to-vector representations trained via neural language models for downstream classification tasks. While Convolutional Neural Networks (CNNs) originated in the field of computer vision, their success in NLP applications—such as semantic parsing, sentence modeling, and query retrieval—is well documented. However, for text-based video classification, the em-

phasis shifts from image and signal-based features to the textual metadata accompanying the video, such as the title and description.

Compared to visual approaches, text-based video categorization offers reduced computational complexity and greater accessibility. Still, it faces limitations when metadata is sparse or absent, and the availability of labeled data remains a challenge.

Despite these constraints, text-based video classification holds considerable promise. It enables efficient categorization using powerful deep learning models that operate solely on textual input [12]. This study exploits both Transformer-based architectures (including BERT, RoBERTa, and ALBERT v2) and recurrent neural networks (specifically LSTM) to learn and generalize the processed textual data. Text classification - defined as the process of assigning textual content to one or more predefined categories - serves as the foundation of this task. Although binary classification is most common, this research tackles a multi-class configuration. The typical workflow involves creating a feature-rich representation of the input text and applying supervised algorithms to determine its correct label. In essence, this study focuses on using deep learning approaches, particularly Transformers and RNNs, for efficient, scalable, and accurate text-based video categorization.

## 1.2 Major Contributions

- Exploratory Data Analysis (EDA) is performed to extract useful insights from video metadata, including daily uploads and category-wise distributions.
- The study proposes a robust text-based video categorization framework using deep learning. Multiple Transformer models, such as BERT, RoBERTa, and ALBERT v2, are fine-tuned on the processed dataset. Additionally, a Recurrent Neural Network (LSTM) is employed for performance comparison.
- Feature engineering is carried out by merging the title and description fields to create a unified column called Processed Text. This consolidated textual input serves as the foundation for model training and classification
- The proposed models are evaluated and compared using key metrics: training and testing accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix analysis

## CHAPTER II

### Methods and materials

#### 2.1 Proposed Methodology

The proposed knowledge mining process for the classification of multiclass text data is shown in Figure 2.1

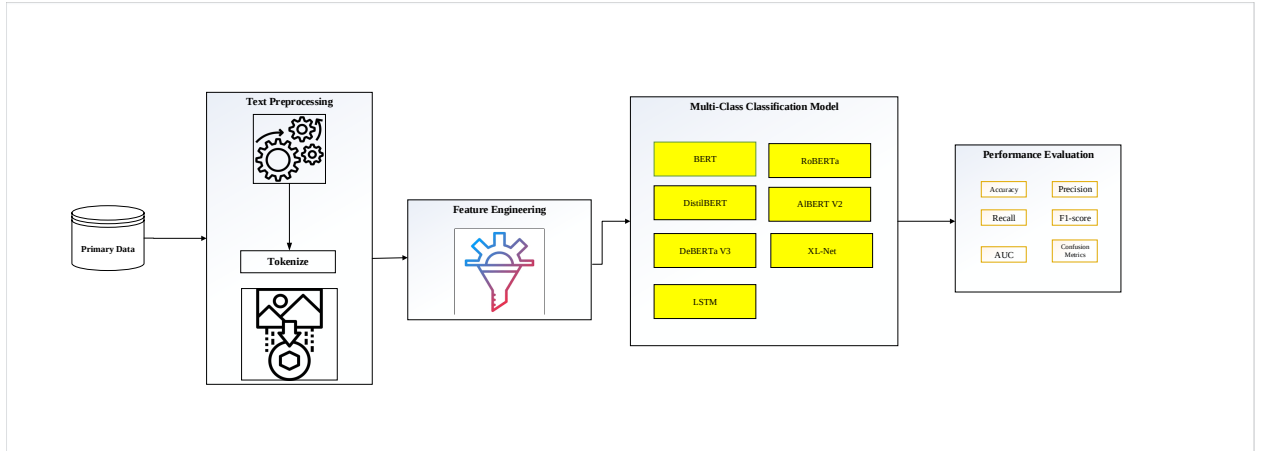


Figure 2.1: Proposed Methodology

- Firstly, a YouTube dataset was collected from publicly available platforms. After collection, several text-cleaning steps were performed, such as converting all words to lowercase and removing punctuation.
- Next, feature engineering was done by merging two columns, namely title and description, to create a single column called Processed Text.
- Then, the dataset was partitioned into training, testing, and validation sets in a 70:20:10 ratio.

- After that, multiple pre-trained transformer models were used, including BERT, DistilBERT, RoBERTa, ALBERT v2, DeBERTa v3, and XLNet. A recurrent neural network model, specifically LSTM, was also trained on the processed data.
- Finally, model performance was evaluated using several well-known metrics such as accuracy, precision, recall, F1 score, AUC, and the Confusion Matrix. Among all, DistilBERT achieved the highest test accuracy of 88%, followed by ALBERT v2 and LSTM.

## 2.2 Model used in Text Classification

1. BERT (Bidirectional Encoder Representations from Transformers) [3] is based on the original Transformer architecture and is pre-trained using two objectives: Masked Language Modeling (MLM) and Next Sentence Prediction (NSP). MLM randomly masks a percentage of input tokens and uses surrounding context to predict them. NSP is a binary classification task where the model determines whether two sentences are consecutive. BERT was trained on the BooksCorpus [13] and English Wikipedia (totaling 16 GB) and achieved state-of-the-art results across eleven NLP tasks, including GLUE, SQuAD, and SWAG.
2. RoBERTa (Robustly Optimized BERT Pretraining Approach) [8] builds on BERT by improving its training procedure. The NSP objective is removed, and the training uses larger batches and longer sequences. RoBERTa is trained on a larger dataset comprising BERT’s corpus, CC-News, OpenWebText, and STORIES amounting to 160 GB of uncompressed text. RoBERTa demonstrated improved performance on benchmarks including GLUE (Wang et al., 2018) and SQuAD (Rajpurkar et al., 2016).
3. DeBERTa (Decoding-enhanced BERT with Disentangled Attention) [5] improves upon BERT and RoBERTa by introducing two key innovations: disentangled attention and enhanced positional encoding. Unlike traditional self-attention mechanisms, DeBERTa separates content and position embeddings, allowing the model to better capture word meaning regardless of token location. Additionally, it uses an improved relative position encoding that enriches contextual understanding during training.
4. XLNet [11] introduces Permuted Language Modeling (PLM), which combines the benefits of autoregressive modeling and MLM. Unlike MLM—which predicts each

masked token independently and introduces a mismatch between training and fine-tuning—PLM predicts each token based on a random permutation of the input, using only preceding tokens in that permutation order. Architecturally, XLNet integrates modifications to the Transformer design, such as two-stream self-attention and target-aware representations. The model is trained on datasets including BooksCorpus, Wikipedia, Giga5 (Graff et al., 2003), ClueWeb 2012-B, and Common Crawl, totaling 158 GB of text.

5. DistilBERT [10] is a smaller, distilled version of BERT that reduces the model size by 40% while retaining approximately 97% of its performance. It uses knowledge distillation (Hinton et al., 2015), where a student model learns to mimic the output distribution of a larger teacher model. DistilBERT employs a triple loss function—combining distillation loss, MLM loss, and cosine embedding loss—and shares similar architecture with fewer layers. It is trained using BERT’s original corpus and is optimized for efficiency.
6. ALBERT (A Lite BERT) [6] is designed to improve scalability and reduce memory usage. It introduces factorized embedding parametrization and cross-layer parameter sharing to lower the total parameter count. In addition to the MLM loss, ALBERT includes a Sentence Order Prediction (SOP) loss, which asks the model to determine whether two sentences are in the correct order or swapped. It is trained on BERT’s dataset along with additional corpora for enhanced performance.
7. LSTM (long-short-term memory) [1] is a type of recurrent neural network designed to capture long-term dependencies in sequential data. Addresses the problem of the disappearance of the gradient commonly found in traditional RNNs by introducing specialized memory cells with gating mechanisms, namely input, forget and output gates, which control the flow of information over time. In this study, LSTM is trained on textual video metadata, where each sequence is limited to 128 tokens and derived from the processed title and description fields. The vocabulary is restricted to the top 10,000 frequent words to reduce dimensionality. LSTM learns temporal relationships in the text and predicts video categories through a softmax output layer.

## 2.3 Performance Metrics

The various assessment parameters used in this study are as follows:

### Accuracy

Represents the proportion of correctly classified instances among all instances.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

### Sensitivity (Recall)

Measures the proportion of actual positives correctly identified.

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

### Specificity (True Negative Rate)

Measures the proportion of actual negatives correctly identified.

$$\text{Specificity} = \frac{TN}{TN + FP}$$

### Type I Error (False Positive Rate)

Occurs when a negative instance is incorrectly classified as positive.

$$\text{Type-I Error} = \frac{FP}{FP + TN}$$

### Type II Error (False Negative Rate)

Occurs when a positive instance is incorrectly classified as negative.

$$\text{Type-II Error} = \frac{FN}{FN + TP}$$

### Precision

Measures the proportion of predicted positives that are actually positive.

$$\text{Precision} = \frac{TP}{TP + FP}$$

### F1-Score

The harmonic mean of precision and recall.

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

### Area Under the ROC Curve (AUROC)

Indicates the model's ability to distinguish between classes.

$$\text{AUROC} = \int_0^1 \text{TPR}(\text{FPR}) d\text{FPR}$$

### Confusion Matrix

A tabular representation of actual vs. predicted classifications. It helps quantify TP, TN, FP, and FN.

## CHAPTER III

# Experiments results & Discussion

### 3.1 Dataset Descriptions

This study utilizes a dataset scraped and created using the YouTube API [4]. The dataset is publicly available on Kaggle [2]. It includes video ID, video title, video description, and category features. Furthermore, it is labeled using the YouTube API, which consists of a YouTube video description and the respective category of the video. The dataset contains 20000 videos falling under nice categories including adventure, art and music, food, history, manufacturing, nature, science and technology, sports, and travel. The category distributions of video category shown in Figure 3.1

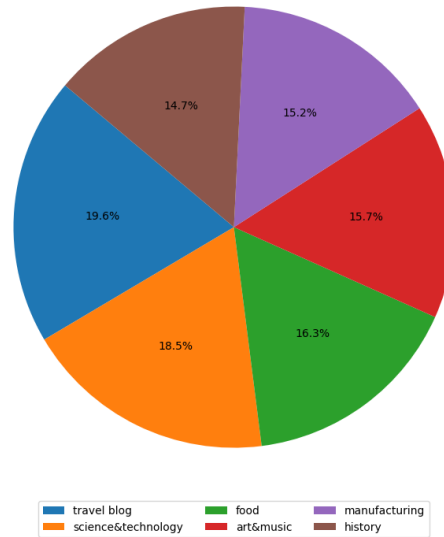


Figure 3.1: Category Distributions of YouTube Video

The Top 50 Word Cloud of Most Frequent Words is depicted in Figure 3.2.



XLM-RoBERTa, designed for multilingual tasks, also demonstrated competitive performance (86.99% accuracy and 0.9769 AUC), suggesting robustness across diverse text patterns.

On the other hand, the LSTM model, a traditional RNN-based architecture, lagged behind with a test accuracy of 83.07% and the lowest AUC of 0.9677. Despite being trained for more epochs, it showed weaker generalization compared to transformer models.

### **3.3 Discussion**

The experimental results demonstrate that transformer-based models consistently outperformed the LSTM model across all evaluation metrics, including accuracy, precision, recall, F1-score, and AUC. Among them, DistilBERT stands out by offering a strong balance between performance and computational efficiency, making it particularly suitable for real-world deployment where resources may be limited. BERT and DeBERTa also delivered high accuracy and robust classification performance, though they require greater computational power. Furthermore, the AUC scores across all transformer models were notably high, indicating their superior ability to distinguish between classes and reinforcing their effectiveness in text classification tasks.

# References

- [1] Xuemei Bai. Text classification based on lstm and attention. In *2018 Thirteenth international conference on digital information management (ICDIM)*, pages 29–32. IEEE, 2018.
- [2] Aashish Bidap. Youtube api data for text categorization. <https://www.kaggle.com/datasets/aashishbidap/youtube-api-data-for-text-categorization>, 2025. Accessed: 2025-07-11.
- [3] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*, pages 4171–4186, 2019.
- [4] Google Developers. Youtube data api v3. <https://developers.google.com/youtube/v3>, 2025. Accessed: 2025-07-11.
- [5] Pengcheng He, Xiaodong Liu, Jianfeng Gao, and Weizhu Chen. Deberta: Decoding-enhanced bert with disentangled attention. *arXiv preprint arXiv:2006.03654*, 2020.
- [6] Zhenzhong Lan, Mingda Chen, Sebastian Goodman, Kevin Gimpel, Piyush Sharma, and Radu Soricut. Albert: A lite bert for self-supervised learning of language representations. *arXiv preprint arXiv:1909.11942*, 2019.
- [7] Feng Li, Jae Chung, and Mark Claypool. Three-year trends in youtube video content and encoding. pages 15–22, 01 2021.
- [8] Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. Roberta: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692*, 2019.

- [9] Daniel W Otter, Julian R Medina, and Jugal K Kalita. A survey of the usages of deep learning for natural language processing. *IEEE transactions on neural networks and learning systems*, 32(2):604–624, 2020.
- [10] Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*, 2019.
- [11] Zhilin Yang, Zihang Dai, Yiming Yang, Jaime Carbonell, Russ R Salakhutdinov, and Quoc V Le. Xlnet: Generalized autoregressive pretraining for language understanding. *Advances in neural information processing systems*, 32, 2019.
- [12] Di Zhang and Mingxing Dong. Improved deep learning model text classification. In *2021 6th International Conference on Intelligent Computing and Signal Processing (ICSP)*, pages 217–220. IEEE, 2021.
- [13] Yukun Zhu, Ryan Kiros, Rich Zemel, Ruslan Salakhutdinov, Raquel Urtasun, Antonio Torralba, and Sanja Fidler. Aligning books and movies: Towards story-like visual explanations by watching movies and reading books. In *Proceedings of the IEEE international conference on computer vision*, pages 19–27, 2015.